Eric Hou

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Education

- Mission San Jose High School (GPA: 4.24), Ohlone College Dual Enrollment
    - **Relevant Courses:** AP Physics C, AP Computer Science A, AP Calc BC, AP Statistics, AP Chemistry, etc
- SAT: 1540/1600 (800 Math, 740 English)

Skills

- Hardware: CAD (Autodesk Fusion 360/Onshape), Manufacturing (3DP, Laser-Cutting, CAM), Electronics (Microcontrollers/Controller Boards/PCB Design)
- Software: Python, HTML, CSS, JS, C++, Familiar with Arduino [Tools/Frameworks: Flask, OpenCV, Git, Docker, React]

Projects (hyweric.github.io/projects)

- Autodesk Fusion 360 Wire Generator [Link]
- Optical Flow Egomotion Localization Algorithm using Computer Vision [Link]
- Flask Applications: Inverse Kinematics Simulator [Link], PortfolioPro [Link], Stick Animator Website [Link]
- Chrome Extensions: Mindful Website Blocker (+Distraction-Free Youtube) [Link]

FTC Team 13356 Hardware 2021 - Present (hyweric.github.io/port/ftc/ftc)

- Accomplishments
    - 2023 and 2024 Northern California Regionals Finalist Alliance Captain
    - Qualified for Maryland Tech Invitational (International Competition of World's Top 40 Teams) over Past 3 Years
    - OPR California #3, Top 30 in World
- Experience
    - Spent 15-30 hours a week designing, manufacturing, testing/collecting data, and integrating HW with SW
    - Things I've Designed (oldest - recent): Extendable Carousel Spinner w/ custom 3DP rails, Suspension Mecanum Drivetrain, Flipping Intake, Dual-Motor Counter Torque Spring Linear Slide Powering Assembly, Stack Intake, Pixel Storage Magazine, Plane Launcher
- Outreach Experience
    - Lesotho Gear Design Inititative - designed custom toys and delivered 100+ to Rural Communities in Lesotho
    - Developed Curriculum and taught CAD and programming to Afghanistan Refugees at the Afghan Coalition
    - Hosted Activities at Tri-Valley Innovation Fair, Santa Clara County Fair, ACSEF, Reid Hillview Airport, Quest Science, Afghan Coalition Health Fair, etc

Student Researcher | ASDRP 2022-2023

- Spent 10+ hours weekly on Deep Learning Project - Using quantum annealing/optimization and a deep learning approach to isolate jets from images of particle collision
- Used Neutral Nets and Regression Algorithms to simulate and filter CERN Data - Experience w/ Docker and ML
- Worked under guiding professor Dr. McMahan, in collaboration with other 5 team members

Head of Computer Science & Member of Board of Directors - Learnify+ Non-Profit Tutoring Organization 2021 - 2023

- Volunteered over 450 hours creating plans and organizing classes for 50+ tutors and 200+ students
- Led the Mission International initiative, expanding the organization's reach to 10 states and 4 countries
- Prepared curriculum for Java, Python, and HTML for other tutors within CS division

USA International Wushu Team 2018 - 2022

- Trained 3x a week (~10 hrs) in Elite Kung Fu Traditional Team
- Represented US in 8th World Kungfu Championships in China for the (Ranked 8 and 18 in 2 categories)
- Other Competitions: Pan-American Wushu, Golden State Wushu Championship, etc

Competitions:

- USACO Gold: Self-taught competitive programming using Python & C++
- **Other:** Toshiba ExploraVision Project Honorable Mention (Total of 6 from Norcal), ASCEF 3rd place, | MLH Hackathon, Google AI Hackathon, and Fremont Hackathon Winner